Data Analytics (DA)

Power BI Assignment 1 – Data Transformation&DataModeling

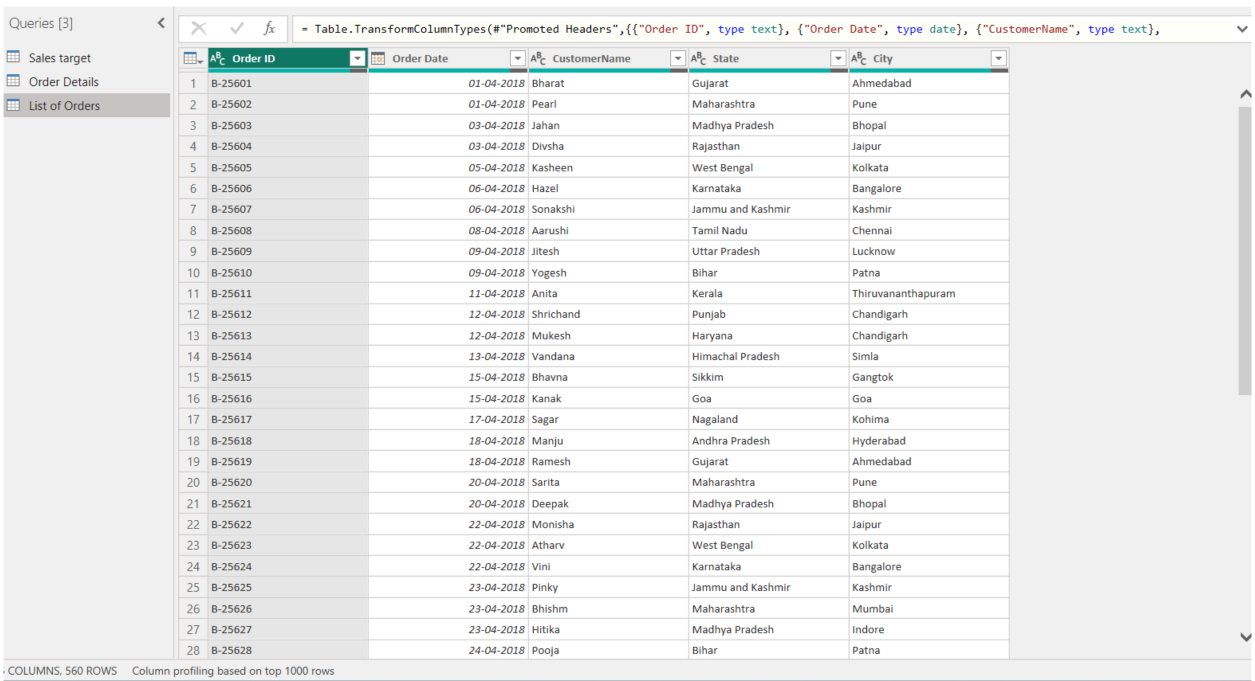
Tasks Data Import & Transformation

1. Data Import:

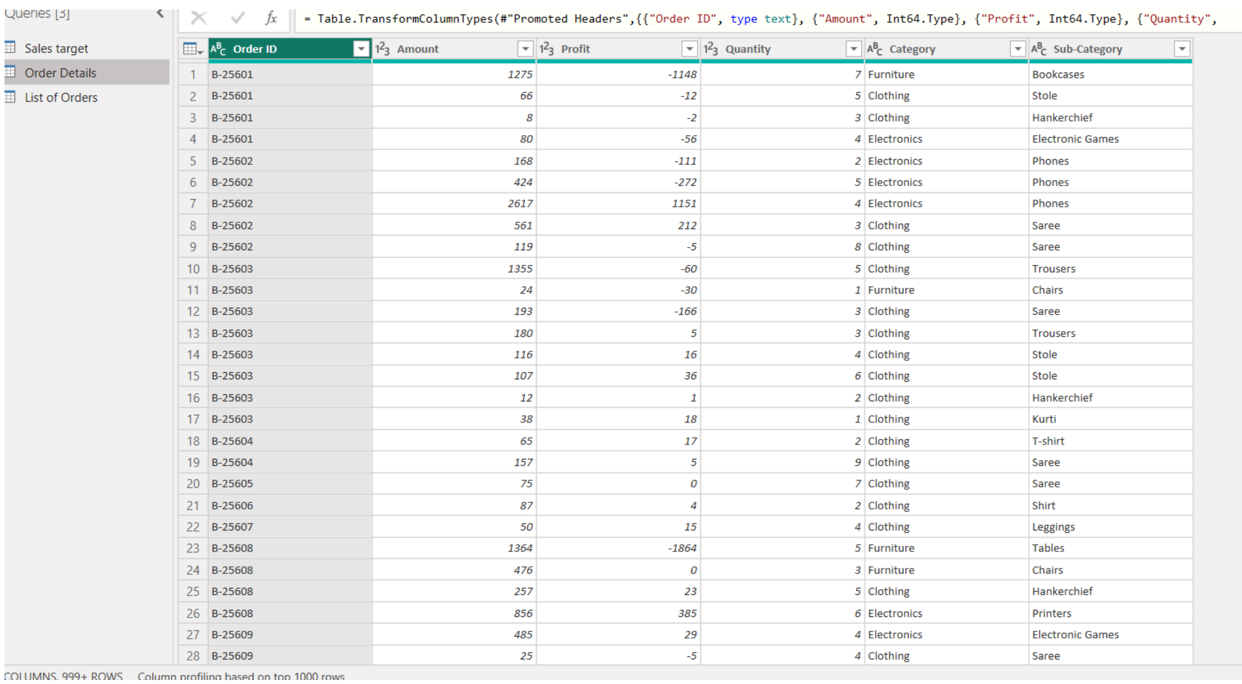
○ Import List of Orders.csv into Power BI.

○ Open the list of orders in the Power Query Editor using the Transform Data option.

○ Import Order Details.csv and Sales Target.csv into Power Query Editor.



	Order ID	Order Date	CustomerName	State	City
1	B-25601	01-04-2018	Bharat	Gujarat	Ahmedabad
2	B-25602	01-04-2018	Pearl	Maharashtra	Pune
3	B-25603	03-04-2018	Jahan	Madhya Pradesh	Bhopal
4	B-25604	03-04-2018	Divsha	Rajasthan	Jaipur
5	B-25605	05-04-2018	Kasheen	West Bengal	Kolkata
6	B-25606	06-04-2018	Hazel	Karnataka	Bangalore
7	B-25607	06-04-2018	Sonakshi	Jammu and Kashmir	Kashmir
8	B-25608	08-04-2018	Aarushi	Tamil Nadu	Chennai
9	B-25609	09-04-2018	Jitesh	Uttar Pradesh	Lucknow
10	B-25610	09-04-2018	Yogesh	Bihar	Patna
11	B-25611	11-04-2018	Anita	Kerala	Thiruvananthapuram
12	B-25612	12-04-2018	Shrichand	Punjab	Chandigarh
13	B-25613	12-04-2018	Mukesh	Haryana	Chandigarh
14	B-25614	13-04-2018	Vandana	Himachal Pradesh	Simla
15	B-25615	15-04-2018	Bhavna	Sikkim	Gangtok
16	B-25616	15-04-2018	Kanak	Goa	Goa
17	B-25617	17-04-2018	Sagar	Nagaland	Kohima
18	B-25618	18-04-2018	Manju	Andhra Pradesh	Hyderabad
19	B-25619	18-04-2018	Ramesh	Gujarat	Ahmedabad
20	B-25620	20-04-2018	Sarita	Maharashtra	Pune
21	B-25621	20-04-2018	Deepak	Madhya Pradesh	Bhopal
22	B-25622	22-04-2018	Monisha	Rajasthan	Jaipur
23	B-25623	22-04-2018	Atharv	West Bengal	Kolkata
24	B-25624	22-04-2018	Vini	Karnataka	Bangalore
25	B-25625	23-04-2018	Pinky	Jammu and Kashmir	Kashmir
26	B-25626	23-04-2018	Bhishm	Maharashtra	Mumbai
27	B-25627	23-04-2018	Hitika	Madhya Pradesh	Indore
28	B-25628	24-04-2018	Pooja	Bihar	Patna



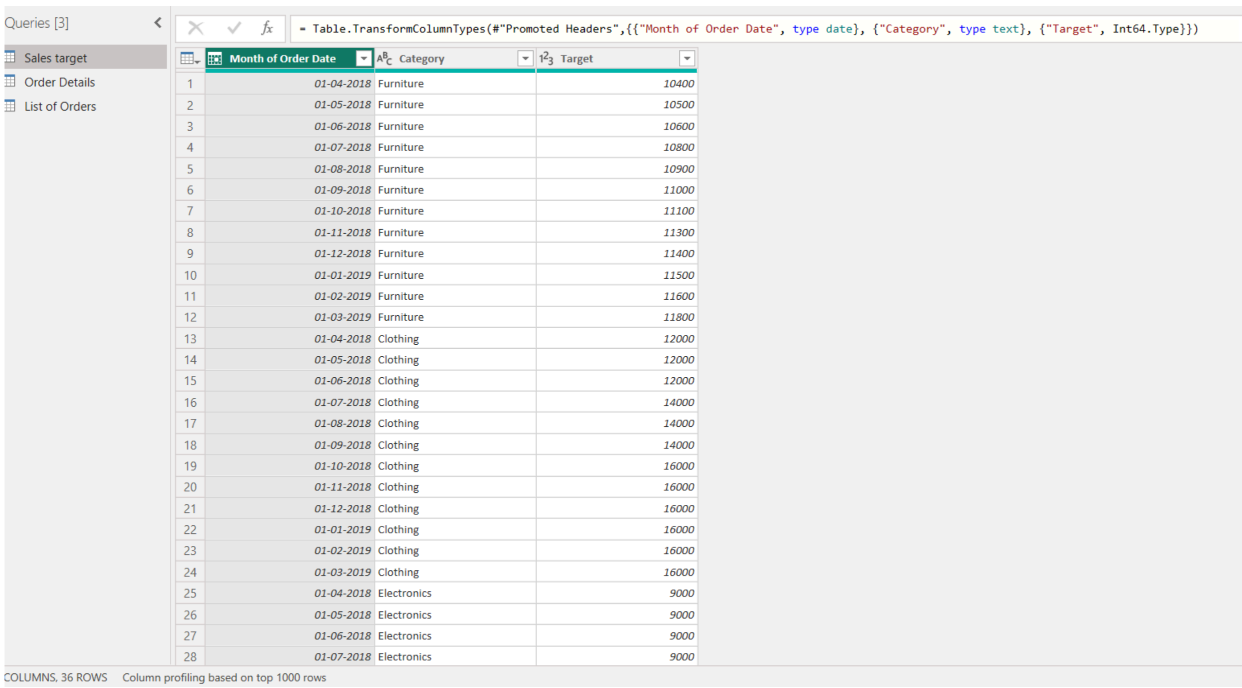
Queries [3]

- Sales target
- Order Details
- List of Orders

= Table.TransformColumnTypes(#"Promoted Headers",{{"Order ID", type text}, {"Amount", Int64.Type}, {"Profit", Int64.Type}, {"Quantity",

	Order ID	Amount	Profit	Quantity	Category	Sub-Category
1	B-25601	1275	-1148	7	Furniture	Bookcases
2	B-25601	66	-12	5	Clothing	Stole
3	B-25601	8	-2	3	Clothing	Hankerchief
4	B-25601	80	-56	4	Electronics	Electronic Games
5	B-25602	168	-111	2	Electronics	Phones
6	B-25602	424	-272	5	Electronics	Phones
7	B-25602	2617	1151	4	Electronics	Phones
8	B-25602	561	212	3	Clothing	Saree
9	B-25602	119	-5	8	Clothing	Saree
10	B-25603	1355	-60	5	Clothing	Trousers
11	B-25603	24	-30	1	Furniture	Chairs
12	B-25603	193	-166	3	Clothing	Saree
13	B-25603	180	5	3	Clothing	Trousers
14	B-25603	116	16	4	Clothing	Stole
15	B-25603	107	36	6	Clothing	Stole
16	B-25603	12	1	2	Clothing	Hankerchief
17	B-25603	38	18	1	Clothing	Kurti
18	B-25604	65	17	2	Clothing	T-shirt
19	B-25604	157	5	9	Clothing	Saree
20	B-25605	75	0	7	Clothing	Saree
21	B-25606	87	4	2	Clothing	Shirt
22	B-25607	50	15	4	Clothing	Leggings
23	B-25608	1364	-1864	5	Furniture	Tables
24	B-25608	476	0	3	Furniture	Chairs
25	B-25608	257	23	5	Clothing	Hankerchief
26	B-25608	856	385	6	Electronics	Printers
27	B-25609	485	29	4	Electronics	Electronic Games
28	B-25609	25	-5	4	Clothing	Saree

COLUMNS, 999+ ROWS Column profiling based on top 1000 rows



Queries [3]

- Sales target
- Order Details
- List of Orders

= Table.TransformColumnTypes(#"Promoted Headers",{{"Month of Order Date", type date}, {"Category", type text}, {"Target", Int64.Type}})

	Month of Order Date	Category	Target
1	01-04-2018	Furniture	10400
2	01-05-2018	Furniture	10500
3	01-06-2018	Furniture	10600
4	01-07-2018	Furniture	10800
5	01-08-2018	Furniture	10900
6	01-09-2018	Furniture	11000
7	01-10-2018	Furniture	11100
8	01-11-2018	Furniture	11300
9	01-12-2018	Furniture	11400
10	01-01-2019	Furniture	11500
11	01-02-2019	Furniture	11600
12	01-03-2019	Furniture	11800
13	01-04-2018	Clothing	12000
14	01-05-2018	Clothing	12000
15	01-06-2018	Clothing	12000
16	01-07-2018	Clothing	14000
17	01-08-2018	Clothing	14000
18	01-09-2018	Clothing	14000
19	01-10-2018	Clothing	16000
20	01-11-2018	Clothing	16000
21	01-12-2018	Clothing	16000
22	01-01-2019	Clothing	16000
23	01-02-2019	Clothing	16000
24	01-03-2019	Clothing	16000
25	01-04-2018	Electronics	9000
26	01-05-2018	Electronics	9000
27	01-06-2018	Electronics	9000
28	01-07-2018	Electronics	9000

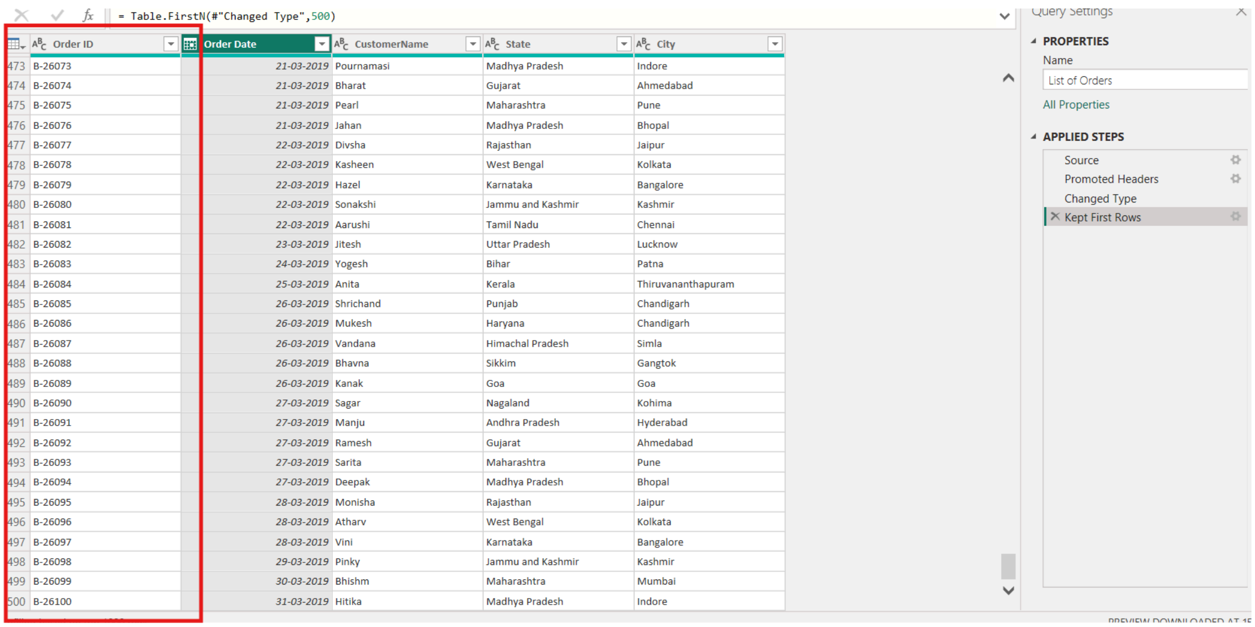
COLUMNS, 36 ROWS Column profiling based on top 1000 rows

1. Successfully imported List of Orders.csv, Order Details.csv, and Sales Target.csv into Power BI.
2. Opened the imported tables in the Power Query Editor using the Transform Data option.
3. Verified that all datasets were loaded correctly for further data transformation.

4. The imported data is ready for cleaning, modeling, and analysis.

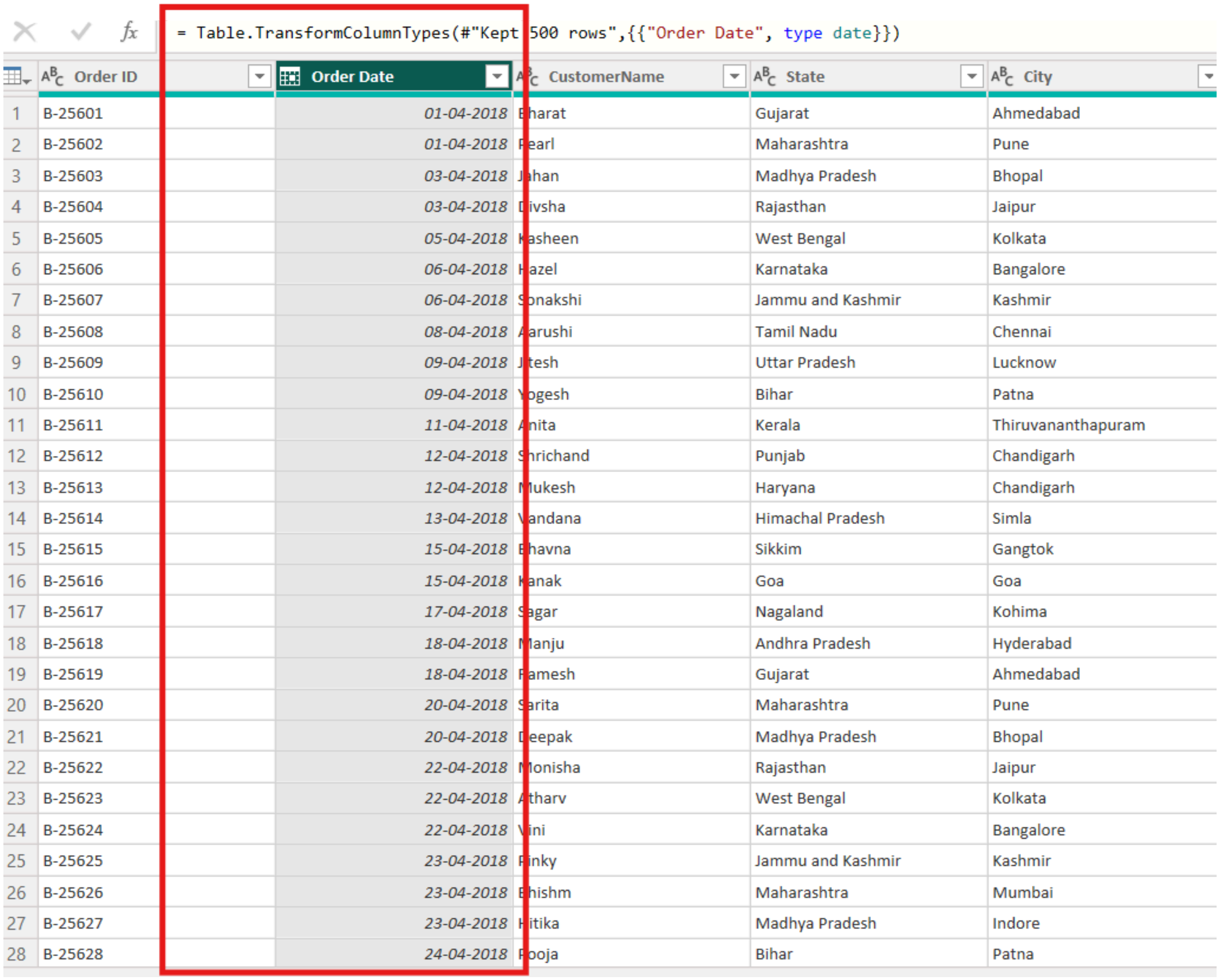
2. Row Limiting & Data Types:

○ Restrict the List of Orders table to the first 500 rows.



	Order ID	Order Date	CustomerName	State	City
473	B-26073	21-03-2019	Pournamasi	Madhya Pradesh	Indore
474	B-26074	21-03-2019	Bharat	Gujarat	Ahmedabad
475	B-26075	21-03-2019	Pearl	Maharashtra	Pune
476	B-26076	21-03-2019	Jahan	Madhya Pradesh	Bhopal
477	B-26077	22-03-2019	Divsha	Rajasthan	Jaipur
478	B-26078	22-03-2019	Kasheen	West Bengal	Kolkata
479	B-26079	22-03-2019	Hazel	Karnataka	Bangalore
480	B-26080	22-03-2019	Sonakshi	Jammu and Kashmir	Kashmir
481	B-26081	22-03-2019	Aarushi	Tamil Nadu	Chennai
482	B-26082	23-03-2019	Jitesh	Uttar Pradesh	Lucknow
483	B-26083	24-03-2019	Yogesh	Bihar	Patna
484	B-26084	25-03-2019	Anita	Kerala	Thiruvananthapuram
485	B-26085	26-03-2019	Shrichand	Punjab	Chandigarh
486	B-26086	26-03-2019	Mukesh	Haryana	Chandigarh
487	B-26087	26-03-2019	Vandana	Himachal Pradesh	Simla
488	B-26088	26-03-2019	Bhavna	Sikkim	Gangtok
489	B-26089	26-03-2019	Kanak	Goa	Goa
490	B-26090	27-03-2019	Sagar	Nagaland	Kohima
491	B-26091	27-03-2019	Manju	Andhra Pradesh	Hyderabad
492	B-26092	27-03-2019	Ramesh	Gujarat	Ahmedabad
493	B-26093	27-03-2019	Sarita	Maharashtra	Pune
494	B-26094	27-03-2019	Deepak	Madhya Pradesh	Bhopal
495	B-26095	28-03-2019	Monisha	Rajasthan	Jaipur
496	B-26096	28-03-2019	Atharv	West Bengal	Kolkata
497	B-26097	28-03-2019	Vini	Karnataka	Bangalore
498	B-26098	29-03-2019	Pinky	Jammu and Kashmir	Kashmir
499	B-26099	30-03-2019	Bhishm	Maharashtra	Mumbai
500	B-26100	31-03-2019	Hitika	Madhya Pradesh	Indore

○ Set the Order Date column to the Date data type.



```
= Table.TransformColumnTypes(#"Kept 500 rows",{{"Order Date", type date}})
```

	Order ID	Order Date	CustomerName	State	City
1	B-25601	01-04-2018	Bharat	Gujarat	Ahmedabad
2	B-25602	01-04-2018	Pearl	Maharashtra	Pune
3	B-25603	03-04-2018	Jahan	Madhya Pradesh	Bhopal
4	B-25604	03-04-2018	Divsha	Rajasthan	Jaipur
5	B-25605	05-04-2018	Kasheen	West Bengal	Kolkata
6	B-25606	06-04-2018	Hazel	Karnataka	Bangalore
7	B-25607	06-04-2018	Sonakshi	Jammu and Kashmir	Kashmir
8	B-25608	08-04-2018	Aarushi	Tamil Nadu	Chennai
9	B-25609	09-04-2018	Jitesh	Uttar Pradesh	Lucknow
10	B-25610	09-04-2018	Yogesh	Bihar	Patna
11	B-25611	11-04-2018	Anita	Kerala	Thiruvananthapuram
12	B-25612	12-04-2018	Shrichand	Punjab	Chandigarh
13	B-25613	12-04-2018	Mukesh	Haryana	Chandigarh
14	B-25614	13-04-2018	Vandana	Himachal Pradesh	Simla
15	B-25615	15-04-2018	Bhavna	Sikkim	Gangtok
16	B-25616	15-04-2018	Kanak	Goa	Goa
17	B-25617	17-04-2018	Sagar	Nagaland	Kohima
18	B-25618	18-04-2018	Manju	Andhra Pradesh	Hyderabad
19	B-25619	18-04-2018	Ramesh	Gujarat	Ahmedabad
20	B-25620	20-04-2018	Sarita	Maharashtra	Pune
21	B-25621	20-04-2018	Deepak	Madhya Pradesh	Bhopal
22	B-25622	22-04-2018	Monisha	Rajasthan	Jaipur
23	B-25623	22-04-2018	Atharv	West Bengal	Kolkata
24	B-25624	22-04-2018	Vini	Karnataka	Bangalore
25	B-25625	23-04-2018	Pinky	Jammu and Kashmir	Kashmir
26	B-25626	23-04-2018	Bhishm	Maharashtra	Mumbai
27	B-25627	23-04-2018	Hitika	Madhya Pradesh	Indore
28	B-25628	24-04-2018	Pooja	Bihar	Patna

○ Change the Amount and Target columns to fixed decimal numbers.

```
= Table.TransformColumnTypes(#"Promoted Headers",{{"Month of Order Date", type date}, {"Category", type text}, {"Target", Currency.Type}})
```

	Month of Order Date	Category	Target
1	01-04-2018	Furniture	10,400.00
2	01-05-2018	Furniture	10,500.00
3	01-06-2018	Furniture	10,600.00
4	01-07-2018	Furniture	10,800.00
5	01-08-2018	Furniture	10,900.00
6	01-09-2018	Furniture	11,000.00
7	01-10-2018	Furniture	11,100.00
8	01-11-2018	Furniture	11,300.00
9	01-12-2018	Furniture	11,400.00
10	01-01-2019	Furniture	11,500.00
11	01-02-2019	Furniture	11,600.00
12	01-03-2019	Furniture	11,800.00
13	01-04-2018	Clothing	12,000.00
14	01-05-2018	Clothing	12,000.00
15	01-06-2018	Clothing	12,000.00
16	01-07-2018	Clothing	14,000.00
17	01-08-2018	Clothing	14,000.00
18	01-09-2018	Clothing	14,000.00
19	01-10-2018	Clothing	16,000.00
20	01-11-2018	Clothing	16,000.00
21	01-12-2018	Clothing	16,000.00
22	01-01-2019	Clothing	16,000.00
23	01-02-2019	Clothing	16,000.00
24	01-03-2019	Clothing	16,000.00
25	01-04-2018	Electronics	9,000.00
26	01-05-2018	Electronics	9,000.00
27	01-06-2018	Electronics	9,000.00
28	01-07-2018	Electronics	9,000.00

```
= Table.TransformColumnTypes(#"Promoted Headers",{{"Order ID", type text}, {"Amount", Currency.Type}, {"Profit", Int64.Type}, {"Quantity",
```

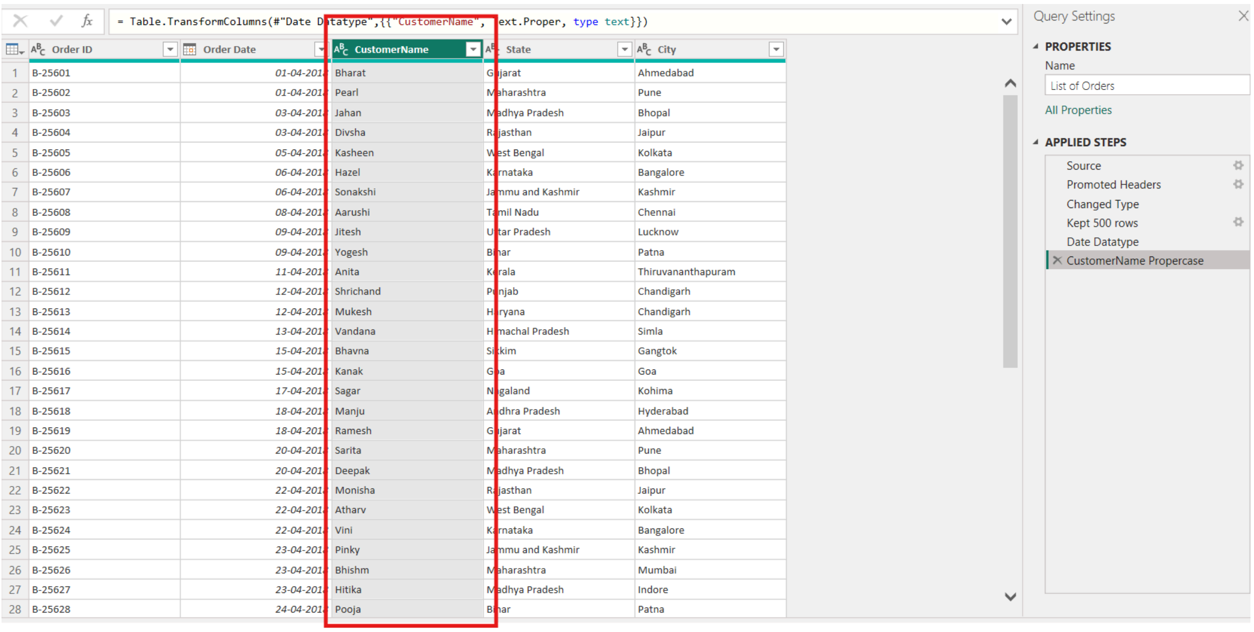
	Order ID	Amount	Profit	Quantity	Category	Sub-Category
1	B-25601	1,275.00	-1148	7	Furniture	Bookcases
2	B-25601	66.00	-12	5	Clothing	Stole
3	B-25601	8.00	-2	3	Clothing	Hankerchief
4	B-25601	80.00	-56	4	Electronics	Electronic Games
5	B-25602	168.00	-111	2	Electronics	Phones
6	B-25602	424.00	-272	5	Electronics	Phones
7	B-25602	2,617.00	1151	4	Electronics	Phones
8	B-25602	561.00	212	3	Clothing	Saree
9	B-25602	119.00	-5	8	Clothing	Saree
10	B-25603	1,355.00	-60	5	Clothing	Trousers
11	B-25603	24.00	-30	1	Furniture	Chairs
12	B-25603	193.00	-166	3	Clothing	Saree
13	B-25603	180.00	5	3	Clothing	Trousers
14	B-25603	116.00	16	4	Clothing	Stole
15	B-25603	107.00	36	6	Clothing	Stole
16	B-25603	12.00	1	2	Clothing	Hankerchief
17	B-25603	38.00	18	1	Clothing	Kurti
18	B-25604	65.00	17	2	Clothing	T-shirt
19	B-25604	157.00	5	9	Clothing	Saree
20	B-25605	75.00	0	7	Clothing	Saree
21	B-25606	87.00	4	2	Clothing	Shirt
22	B-25607	50.00	15	4	Clothing	Leggings
23	B-25608	1,364.00	-1864	5	Furniture	Tables
24	B-25608	476.00	0	3	Furniture	Chairs
25	B-25608	257.00	23	5	Clothing	Hankerchief
26	B-25608	856.00	385	6	Electronics	Printers
27	B-25609	485.00	29	4	Electronics	Electronic Games
28	B-25609	25.00	-5	4	Clothing	Saree

1.The List of Orders table was limited to the first 500 rows for focused analysis.
2.The Order Date column was converted to the Date data type.

3.The Amount and Target columns were changed to Fixed Decimal Number data types.
4.Correct data types improve calculation accuracy and data consistency.

3. Text Formatting:

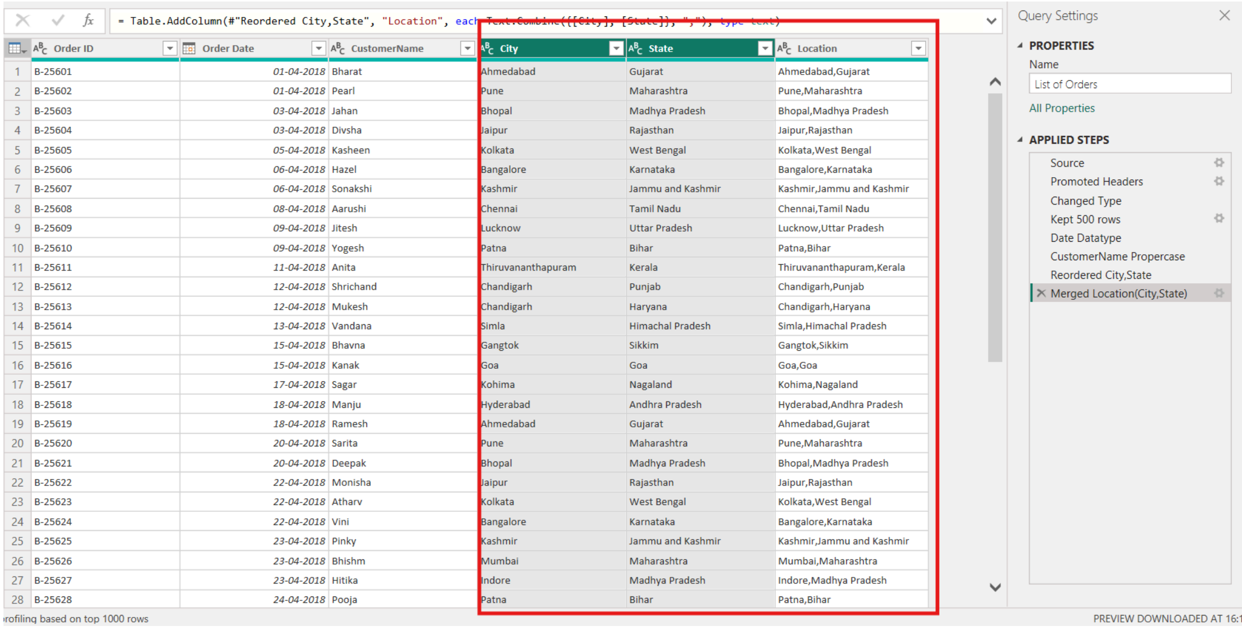
○ Format the CustomerName column to proper case to ensure consistent capitalization**.**



1.The CustomerName column was formatted to Proper Case for consistent capitalization.
2.All customer names now follow a uniform and professional text format.

4. Column Creation:

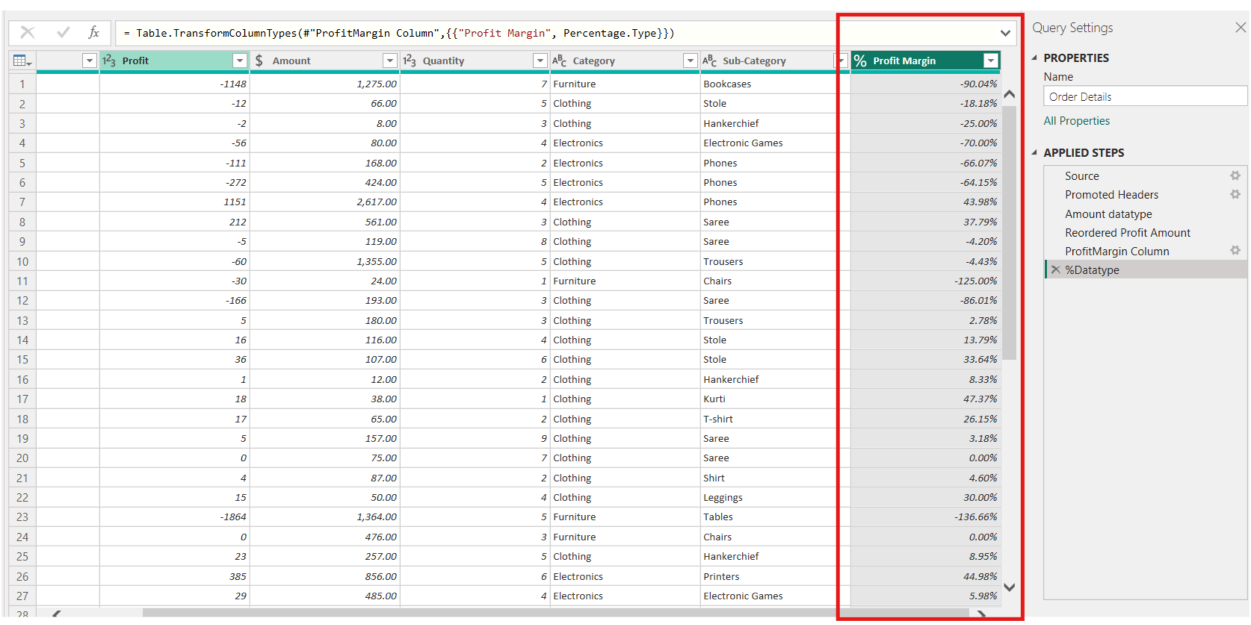
○ Merge the City and State columns to create a new column named Location in the format: City, State



```
= Table.AddColumn(#"Reordered City,State", "Location", each Text.Combine({[City], [State]}, ","), type text)
```

	Order ID	Order Date	CustomerName	City	State	Location
1	B-25601	01-04-2018	Bharat	Ahmedabad	Gujarat	Ahmedabad,Gujarat
2	B-25602	01-04-2018	Pearl	Pune	Maharashtra	Pune,Maharashtra
3	B-25603	03-04-2018	Jahan	Bhopal	Madhya Pradesh	Bhopal,Madhya Pradesh
4	B-25604	03-04-2018	Divsha	Jaipur	Rajasthan	Jaipur,Rajasthan
5	B-25605	05-04-2018	Kasheen	Kolkata	West Bengal	Kolkata,West Bengal
6	B-25606	06-04-2018	Hazel	Bangalore	Karnataka	Bangalore,Karnataka
7	B-25607	06-04-2018	Sonakshi	Kashmir	Jammu and Kashmir	Kashmir,Jammu and Kashmir
8	B-25608	08-04-2018	Aarushi	Chennai	Tamil Nadu	Chennai,Tamil Nadu
9	B-25609	09-04-2018	Jitesh	Lucknow	Uttar Pradesh	Lucknow,Uttar Pradesh
10	B-25610	09-04-2018	Yogesh	Patna	Bihar	Patna,Bihar
11	B-25611	11-04-2018	Anita	Thiruvananthapuram	Kerala	Thiruvananthapuram,Kerala
12	B-25612	12-04-2018	Shrichand	Chandigarh	Punjab	Chandigarh,Punjab
13	B-25613	12-04-2018	Mukesh	Chandigarh	Haryana	Chandigarh,Haryana
14	B-25614	13-04-2018	Vandana	Simla	Himachal Pradesh	Simla,Himachal Pradesh
15	B-25615	15-04-2018	Bhavna	Gangtok	Sikkim	Gangtok,Sikkim
16	B-25616	15-04-2018	Kanak	Goa	Goa	Goa,Goa
17	B-25617	17-04-2018	Sagar	Kohima	Nagaland	Kohima,Nagaland
18	B-25618	18-04-2018	Manju	Hyderabad	Andhra Pradesh	Hyderabad,Andhra Pradesh
19	B-25619	18-04-2018	Ramesh	Ahmedabad	Gujarat	Ahmedabad,Gujarat
20	B-25620	20-04-2018	Sarita	Pune	Maharashtra	Pune,Maharashtra
21	B-25621	20-04-2018	Deepak	Bhopal	Madhya Pradesh	Bhopal,Madhya Pradesh
22	B-25622	22-04-2018	Monisha	Jaipur	Rajasthan	Jaipur,Rajasthan
23	B-25623	22-04-2018	Atharv	Kolkata	West Bengal	Kolkata,West Bengal
24	B-25624	22-04-2018	Vini	Bangalore	Karnataka	Bangalore,Karnataka
25	B-25625	23-04-2018	Pinky	Kashmir	Jammu and Kashmir	Kashmir,Jammu and Kashmir
26	B-25626	23-04-2018	Bhishm	Mumbai	Maharashtra	Mumbai,Maharashtra
27	B-25627	23-04-2018	Hitika	Indore	Madhya Pradesh	Indore,Madhya Pradesh
28	B-25628	24-04-2018	Pooja	Patna	Bihar	Patna,Bihar

○ Create a custom column named Profit Margin calculated as: Profit / Amount (expressed as a percentage)



```
= Table.TransformColumnTypes(#"ProfitMargin Column",{{"Profit Margin", Percentage.Type}})
```

		Profit	Amount	Quantity	Category	Sub-Category	Profit Margin
1		-1148	1,275.00	7	Furniture	Bookcases	-90.04%
2		-12	66.00	5	Clothing	Stole	-18.18%
3		-2	8.00	3	Clothing	Hankerchief	-25.00%
4		-56	80.00	4	Electronics	Electronic Games	-70.00%
5		-111	168.00	2	Electronics	Phones	-66.07%
6		-272	424.00	5	Electronics	Phones	-64.15%
7		1151	2,617.00	4	Electronics	Phones	43.98%
8		212	561.00	3	Clothing	Saree	37.79%
9		-5	119.00	8	Clothing	Saree	-4.20%
10		-60	1,355.00	5	Clothing	Trousers	-4.43%
11		-30	24.00	1	Furniture	Chairs	-125.00%
12		-166	193.00	3	Clothing	Saree	-86.01%
13		5	180.00	3	Clothing	Trousers	2.78%
14		16	116.00	4	Clothing	Stole	13.79%
15		36	107.00	6	Clothing	Stole	33.64%
16		1	12.00	2	Clothing	Hankerchief	8.33%
17		18	38.00	1	Clothing	Kurti	47.37%
18		17	65.00	2	Clothing	T-shirt	26.15%
19		5	157.00	9	Clothing	Saree	3.18%
20		0	75.00	7	Clothing	Saree	0.00%
21		4	87.00	2	Clothing	Shirt	4.60%
22		15	50.00	4	Clothing	Leggings	30.00%
23		-1864	1,364.00	5	Furniture	Tables	-136.66%
24		0	476.00	3	Furniture	Chairs	0.00%
25		23	257.00	5	Clothing	Hankerchief	8.95%
26		385	856.00	6	Electronics	Printers	44.98%
27		29	485.00	4	Electronics	Electronic Games	5.98%

1. The City and State columns were merged to create a new Location column in the format City, State.
2. A custom column Profit Margin was created by calculating Profit ÷ Amount and formatting the result as a percentage.
3. The Location column provides a clear combined address field for analysis.
4. The Profit Margin column helps evaluate profitability for each order.
5. These new columns improve data analysis and reporting.

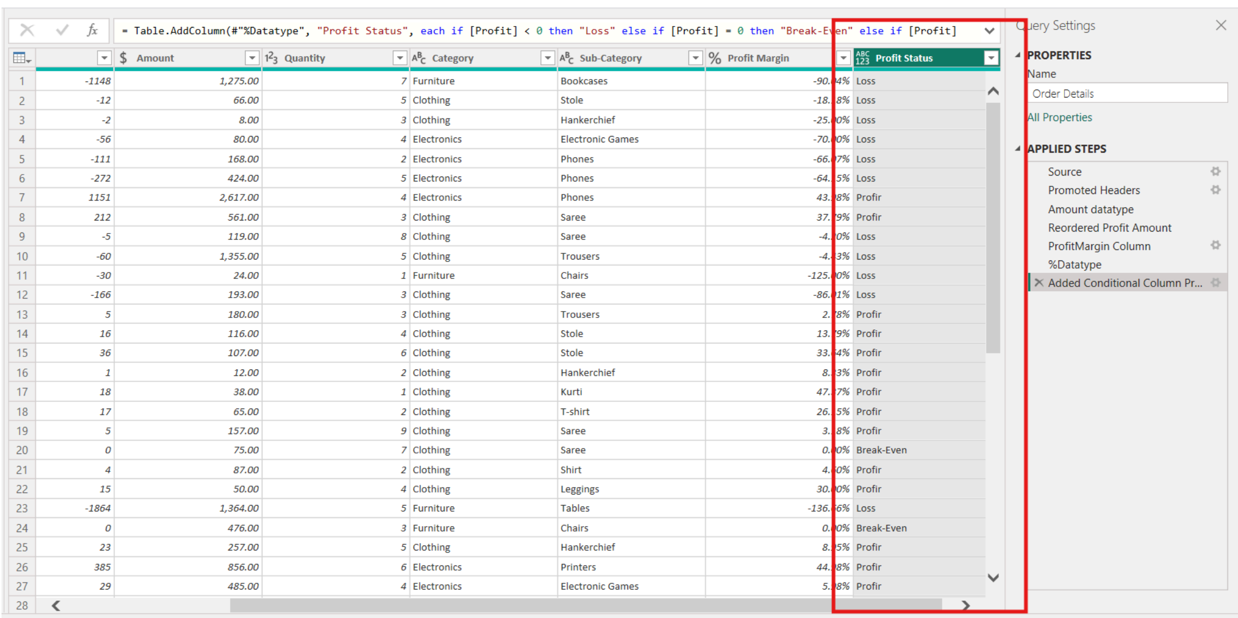
5. Conditional Column:

○ Add a conditional column named Profit Status based on:

■ Profit < 0 → Loss

■ Profit = 0 → Break-Even

■ Profit > 0 → Profit



1.A conditional column named Profit Status was created based on the Profit values.

Orders with Profit < 0 were classified as Loss.

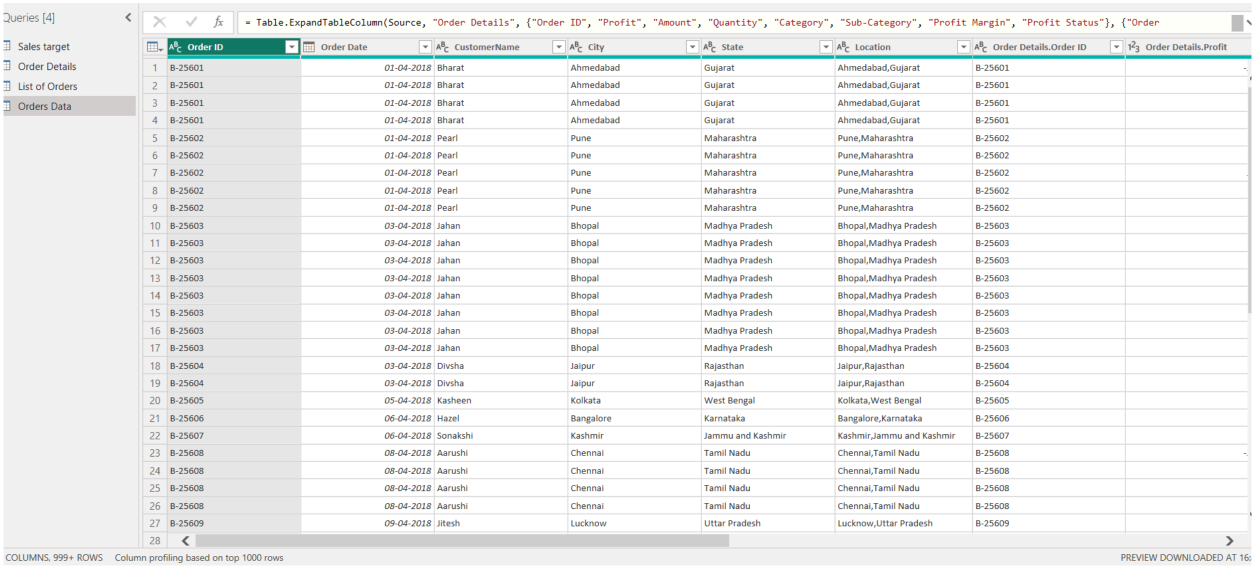
Orders with Profit = 0 were classified as Break-Even.

Orders with Profit > 0 were classified as Profit.

2.The new column makes it easier to analyze profitable, loss-making, and break-even orders.

6. Merging Data (Joins)

○ Merge the List of Orders and Order Details tables using Order ID.
○ Name the merged output table as Orders Data.

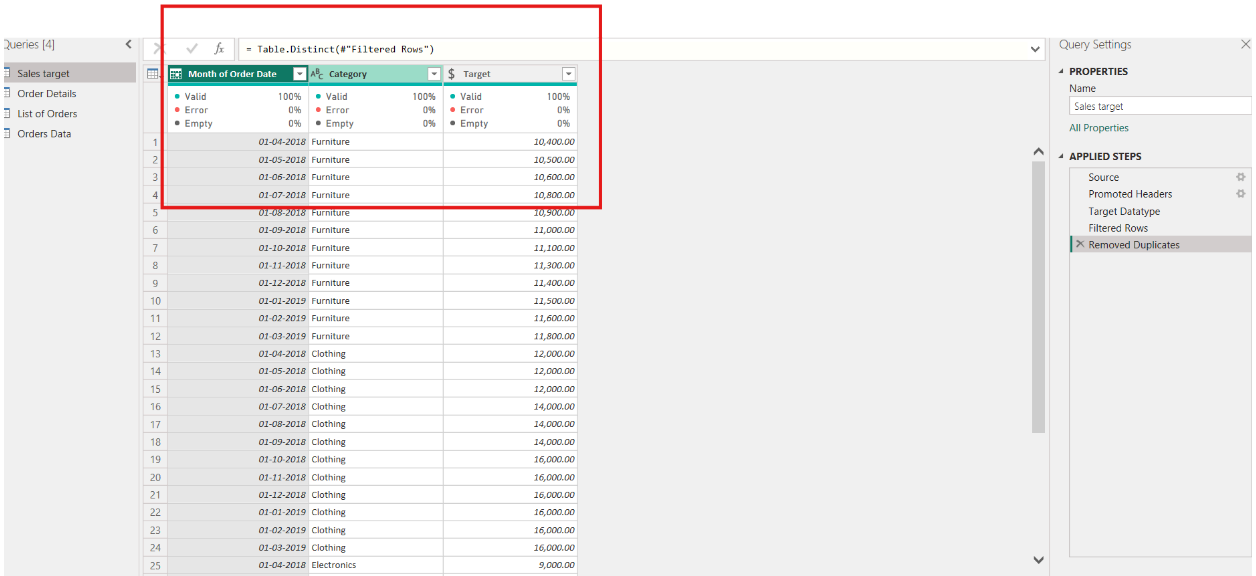


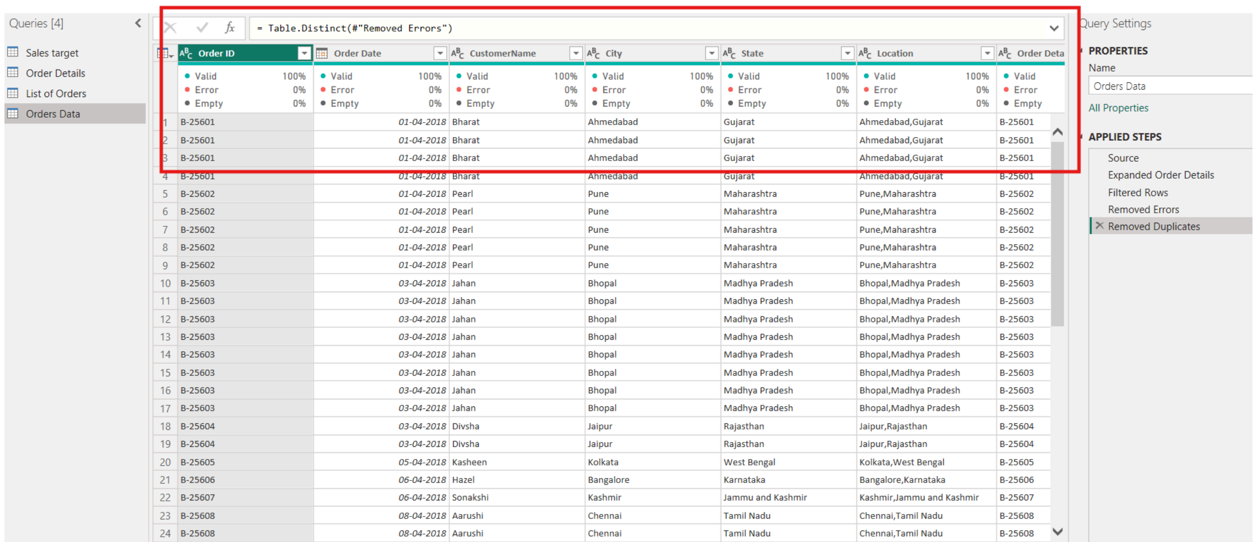
	Order ID	Order Date	CustomerName	City	State	Location	Order Details.Order ID	Order Details.Profit
1	B-25601	01-04-2018	Bharat	Ahmedabad	Gujarat	Ahmedabad,Gujarat	B-25601	
2	B-25601	01-04-2018	Bharat	Ahmedabad	Gujarat	Ahmedabad,Gujarat	B-25601	
3	B-25601	01-04-2018	Bharat	Ahmedabad	Gujarat	Ahmedabad,Gujarat	B-25601	
4	B-25601	01-04-2018	Bharat	Ahmedabad	Gujarat	Ahmedabad,Gujarat	B-25601	
5	B-25602	01-04-2018	Pearl	Pune	Maharashtra	Pune,Maharashtra	B-25602	
6	B-25602	01-04-2018	Pearl	Pune	Maharashtra	Pune,Maharashtra	B-25602	
7	B-25602	01-04-2018	Pearl	Pune	Maharashtra	Pune,Maharashtra	B-25602	
8	B-25602	01-04-2018	Pearl	Pune	Maharashtra	Pune,Maharashtra	B-25602	
9	B-25602	01-04-2018	Pearl	Pune	Maharashtra	Pune,Maharashtra	B-25602	
10	B-25603	03-04-2018	Jahan	Bhopal	Madhya Pradesh	Bhopal,Madhya Pradesh	B-25603	
11	B-25603	03-04-2018	Jahan	Bhopal	Madhya Pradesh	Bhopal,Madhya Pradesh	B-25603	
12	B-25603	03-04-2018	Jahan	Bhopal	Madhya Pradesh	Bhopal,Madhya Pradesh	B-25603	
13	B-25603	03-04-2018	Jahan	Bhopal	Madhya Pradesh	Bhopal,Madhya Pradesh	B-25603	
14	B-25603	03-04-2018	Jahan	Bhopal	Madhya Pradesh	Bhopal,Madhya Pradesh	B-25603	
15	B-25603	03-04-2018	Jahan	Bhopal	Madhya Pradesh	Bhopal,Madhya Pradesh	B-25603	
16	B-25603	03-04-2018	Jahan	Bhopal	Madhya Pradesh	Bhopal,Madhya Pradesh	B-25603	
17	B-25603	03-04-2018	Jahan	Bhopal	Madhya Pradesh	Bhopal,Madhya Pradesh	B-25603	
18	B-25604	03-04-2018	Divsha	Jaipur	Rajasthan	Jaipur,Rajasthan	B-25604	
19	B-25604	03-04-2018	Divsha	Jaipur	Rajasthan	Jaipur,Rajasthan	B-25604	
20	B-25605	05-04-2018	Kasheen	Kolkata	West Bengal	Kolkata,West Bengal	B-25605	
21	B-25606	06-04-2018	Hazel	Bangalore	Karnataka	Bangalore,Karnataka	B-25606	
22	B-25607	06-04-2018	Sonakshi	Kashmir	Jammu and Kashmir	Kashmir,Jammu and Kashmir	B-25607	
23	B-25608	08-04-2018	Aarushi	Chennai	Tamil Nadu	Chennai,Tamil Nadu	B-25608	
24	B-25608	08-04-2018	Aarushi	Chennai	Tamil Nadu	Chennai,Tamil Nadu	B-25608	
25	B-25608	08-04-2018	Aarushi	Chennai	Tamil Nadu	Chennai,Tamil Nadu	B-25608	
26	B-25608	08-04-2018	Aarushi	Chennai	Tamil Nadu	Chennai,Tamil Nadu	B-25608	
27	B-25609	09-04-2018	Jitesh	Lucknow	Uttar Pradesh	Lucknow,Uttar Pradesh	B-25609	

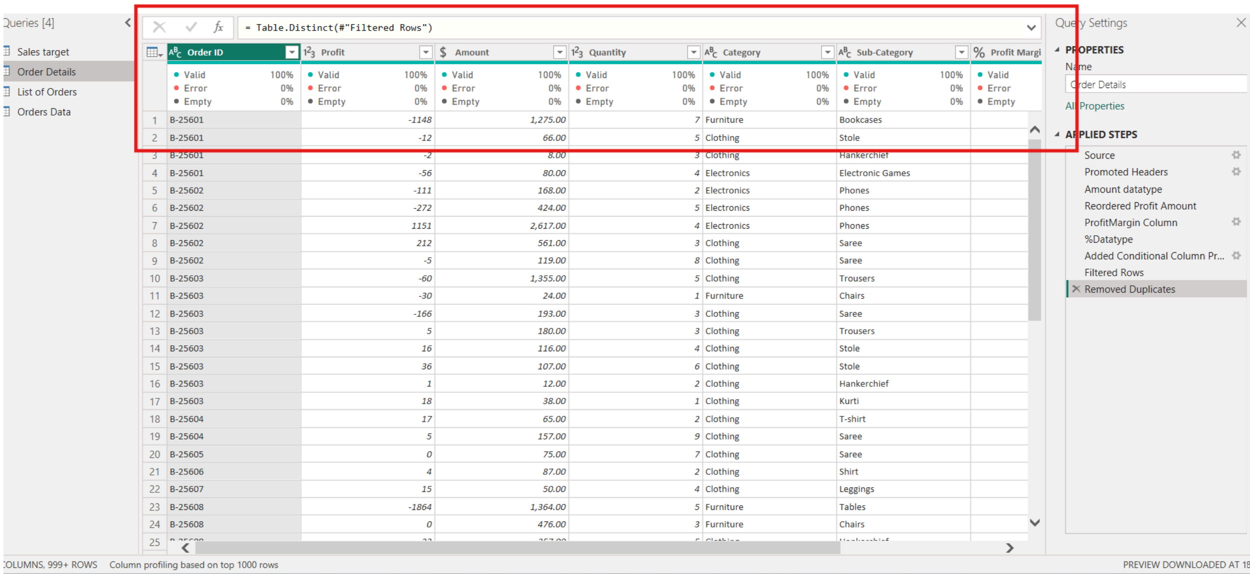
1. The List of Orders and Order Details tables were successfully merged using Order ID.
2. The merged table was named Orders Data.
3. The merge combined related information from both tables into a single dataset.
4. The merged data is ready for further analysis.

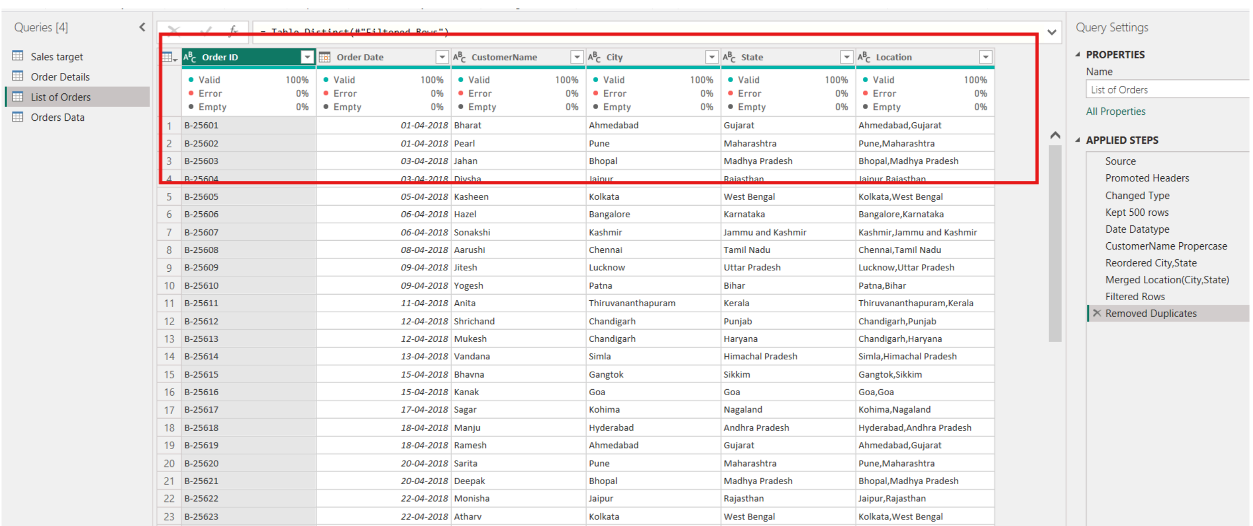
7. Handling Missing Data & Duplicate Data

- Identify missing values across all tables.
- Define and apply appropriate strategies such as:
  - ○ Replacing missing values
  - ○ Removing records
  - ○ Retaining nulls with justification
- Identify duplicate rows and define a suitable handling strategy based on business logic.


Queries [4]
Sales target
Order Details
List of Orders
Orders Data
= Table.Distinct(#"Filtered Rows")
Month of Order Date
Category
Target
Valid 100%
Error 0%
Empty 0%
1 01-04-2018 Furniture 10,400.00
2 01-05-2018 Furniture 10,500.00
3 01-06-2018 Furniture 10,600.00
4 01-07-2018 Furniture 10,800.00
5 01-08-2018 Furniture 10,900.00
6 01-09-2018 Furniture 11,000.00
7 01-10-2018 Furniture 11,100.00
8 01-11-2018 Furniture 11,300.00
9 01-12-2018 Furniture 11,400.00
10 01-01-2019 Furniture 11,500.00
11 01-02-2019 Furniture 11,600.00
12 01-03-2019 Furniture 11,800.00
13 01-04-2018 Clothing 12,000.00
14 01-05-2018 Clothing 12,000.00
15 01-06-2018 Clothing 12,000.00
16 01-07-2018 Clothing 14,000.00
17 01-08-2018 Clothing 14,000.00
18 01-09-2018 Clothing 14,000.00
19 01-10-2018 Clothing 16,000.00
20 01-11-2018 Clothing 16,000.00
21 01-12-2018 Clothing 16,000.00
22 01-01-2019 Clothing 16,000.00
23 01-02-2019 Clothing 16,000.00
24 01-03-2019 Clothing 16,000.00
25 01-04-2018 Electronics 9,000.00
Query Settings
PROPERTIES
Name
Sales target
All Properties
APPLIED STEPS
Source
Promoted Headers
Target Datatype
Filtered Rows
Removed Duplicates


Queries [4]
Sales target
Order Details
List of Orders
Orders Data
= Table.Distinct(#"Removed Errors")
Order ID
Order Date
CustomerName
City
State
Location
Order Deta
Valid 100%
Error 0%
Empty 0%
1 B-25601 01-04-2018 Bharat Ahmedabad Gujarat Ahmedabad,Gujarat B-25601
2 B-25601 01-04-2018 Bharat Ahmedabad Gujarat Ahmedabad,Gujarat B-25601
3 B-25601 01-04-2018 Bharat Ahmedabad Gujarat Ahmedabad,Gujarat B-25601
4 B-25601 01-04-2018 Bharat Ahmedabad Gujarat Ahmedabad,Gujarat B-25601
5 B-25602 01-04-2018 Pearl Pune Maharashtra Pune,Maharashtra B-25602
6 B-25602 01-04-2018 Pearl Pune Maharashtra Pune,Maharashtra B-25602
7 B-25602 01-04-2018 Pearl Pune Maharashtra Pune,Maharashtra B-25602
8 B-25602 01-04-2018 Pearl Pune Maharashtra Pune,Maharashtra B-25602
9 B-25602 01-04-2018 Pearl Pune Maharashtra Pune,Maharashtra B-25602
10 B-25603 03-04-2018 Jahan Bhopal Madhya Pradesh Bhopal,Madhya Pradesh B-25603
11 B-25603 03-04-2018 Jahan Bhopal Madhya Pradesh Bhopal,Madhya Pradesh B-25603
12 B-25603 03-04-2018 Jahan Bhopal Madhya Pradesh Bhopal,Madhya Pradesh B-25603
13 B-25603 03-04-2018 Jahan Bhopal Madhya Pradesh Bhopal,Madhya Pradesh B-25603
14 B-25603 03-04-2018 Jahan Bhopal Madhya Pradesh Bhopal,Madhya Pradesh B-25603
15 B-25603 03-04-2018 Jahan Bhopal Madhya Pradesh Bhopal,Madhya Pradesh B-25603
16 B-25603 03-04-2018 Jahan Bhopal Madhya Pradesh Bhopal,Madhya Pradesh B-25603
17 B-25603 03-04-2018 Jahan Bhopal Madhya Pradesh Bhopal,Madhya Pradesh B-25603
18 B-25604 03-04-2018 Divsha Jaipur Rajasthan Jaipur,Rajasthan B-25604
19 B-25604 03-04-2018 Divsha Jaipur Rajasthan Jaipur,Rajasthan B-25604
20 B-25605 05-04-2018 Kasheen Kolkata West Bengal Kolkata,West Bengal B-25605
21 B-25606 06-04-2018 Hazel Bangalore Karnataka Bangalore,Karnataka B-25606
22 B-25607 06-04-2018 Sonakshi Kashmir Jammu and Kashmir Kashmir,Jammu and Kashmir B-25607
23 B-25608 08-04-2018 Aarushi Chennai Tamil Nadu Chennai,Tamil Nadu B-25608
24 B-25608 08-04-2018 Aarushi Chennai Tamil Nadu Chennai,Tamil Nadu B-25608
Query Settings
PROPERTIES
Name
Orders Data
All Properties
APPLIED STEPS
Source
Expanded Order Details
Filtered Rows
Removed Errors
Removed Duplicates

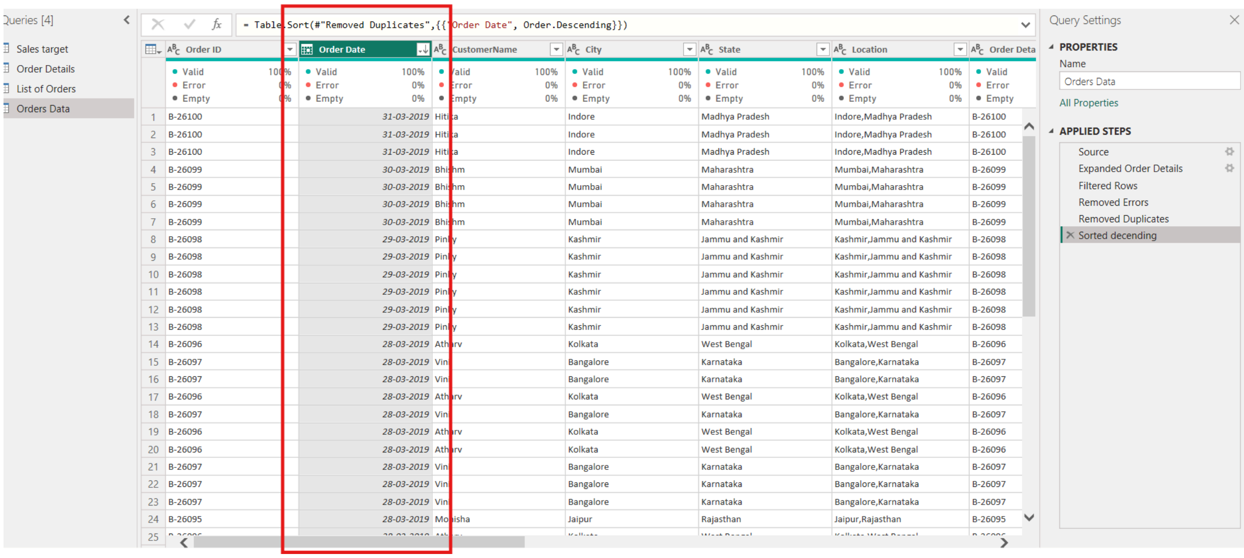




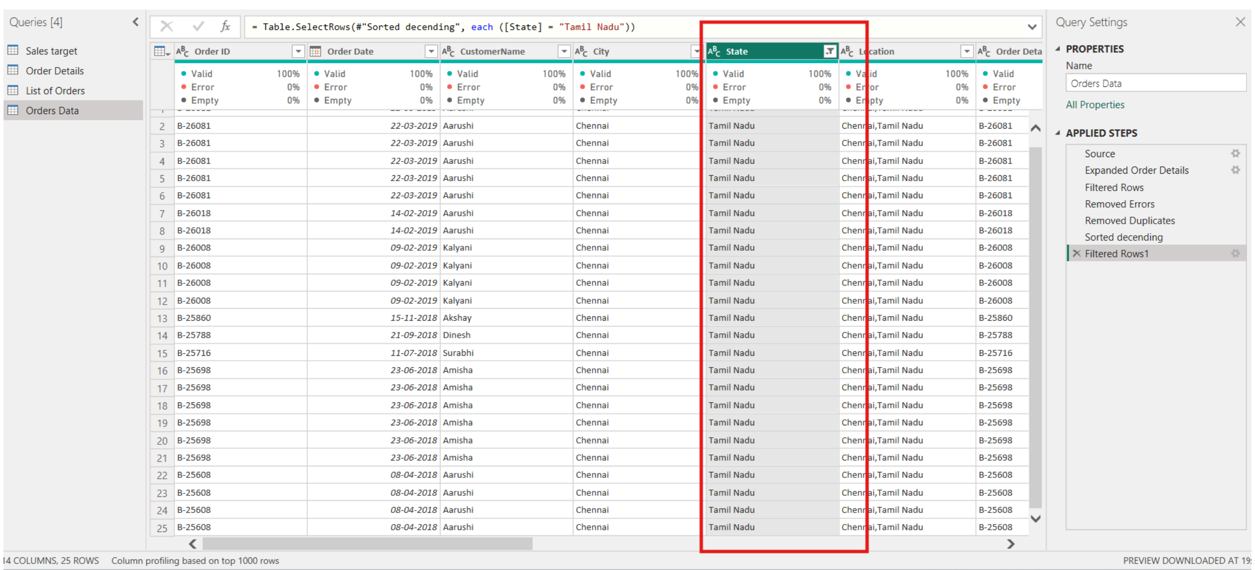
1. Missing values were identified across all tables.
2. Appropriate methods such as replacing missing values, removing unnecessary records, or retaining nulls with valid justification were applied.
3. Data quality was improved while preserving important information.
4. Duplicate rows were identified and checked.

8. Sorting and Filtering Data

● Sort records in the Orders Data table by Order Date in descending order.



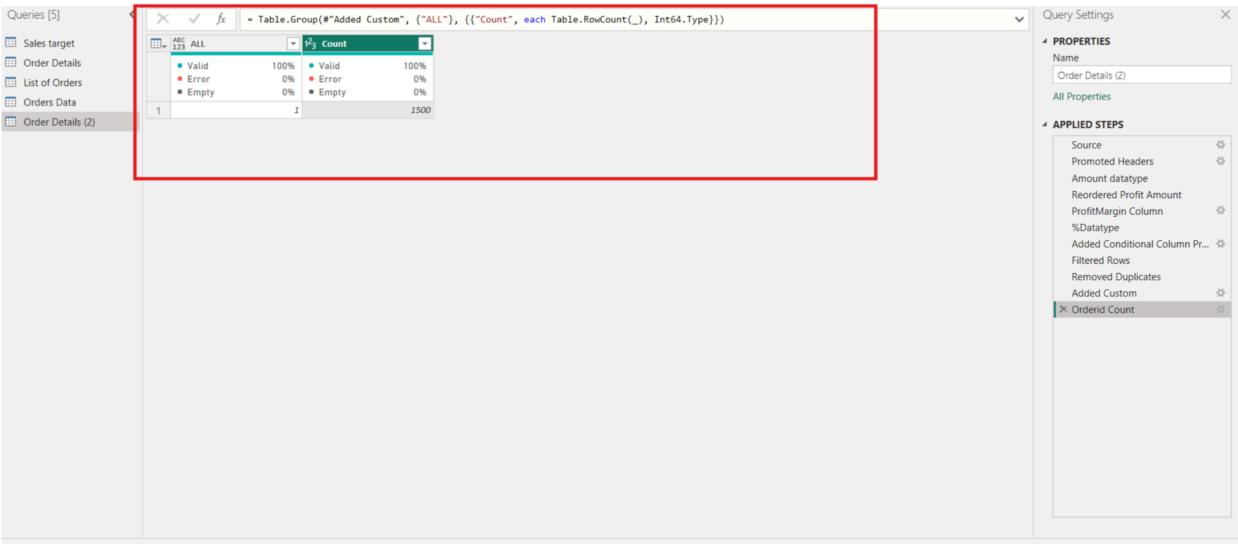
- Apply filters to analyze orders from a specific state (e.g., Tamil Nadu).



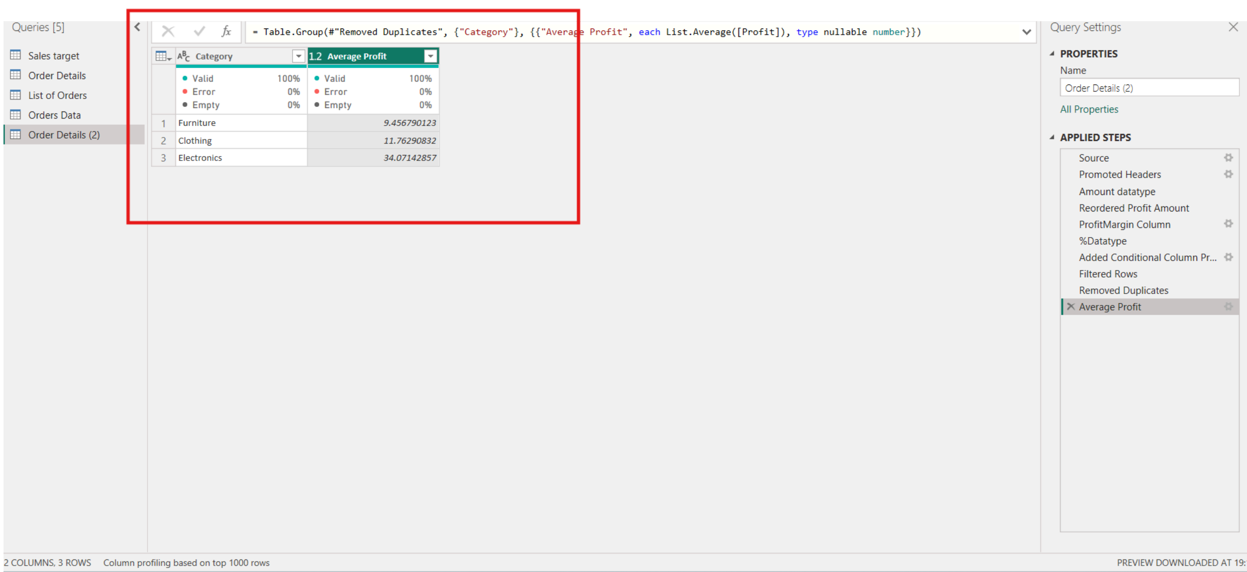
1. The Orders Data table was sorted by Order Date in descending order to display the most recent orders first.
2. Filters were applied to view orders from the selected state (Tamil Nadu).
3. The sorted and filtered data made it easier to identify recent orders and analyze regional sales.

9. Grouping and Aggregating Data

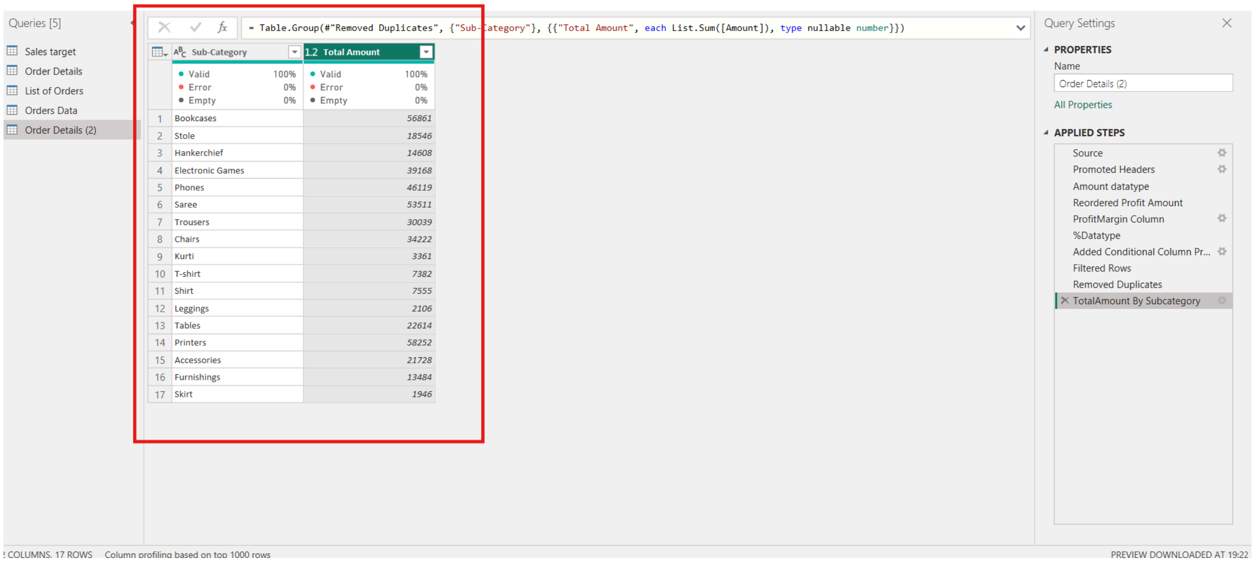
- Duplicate the Order Details table and perform the following:
    - Count of Order ID



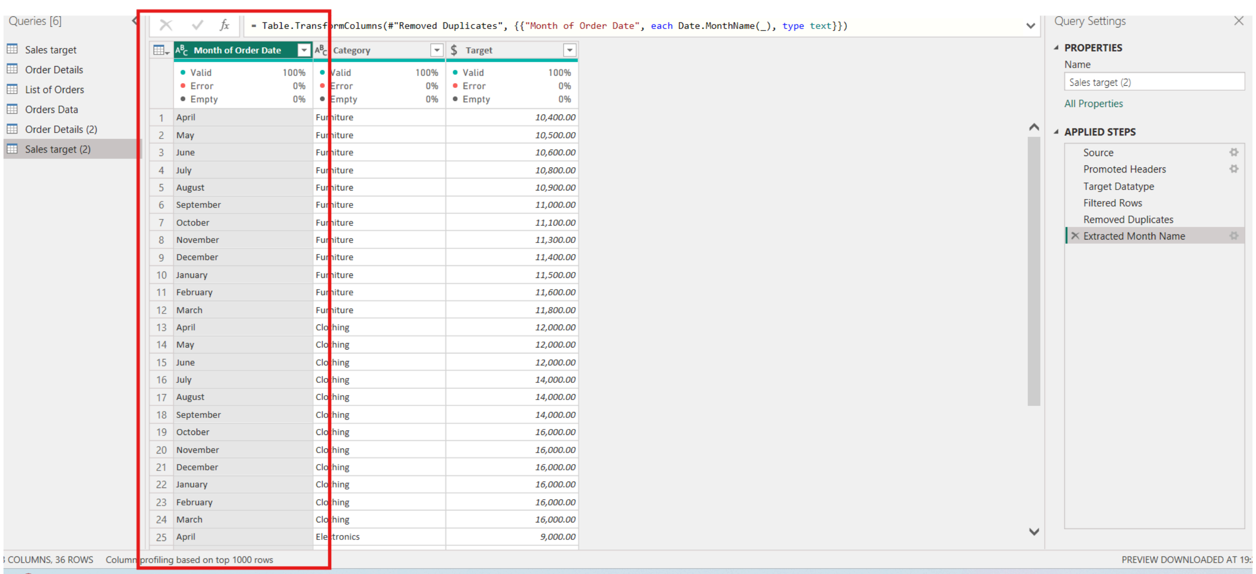
○ Average Profit by Category

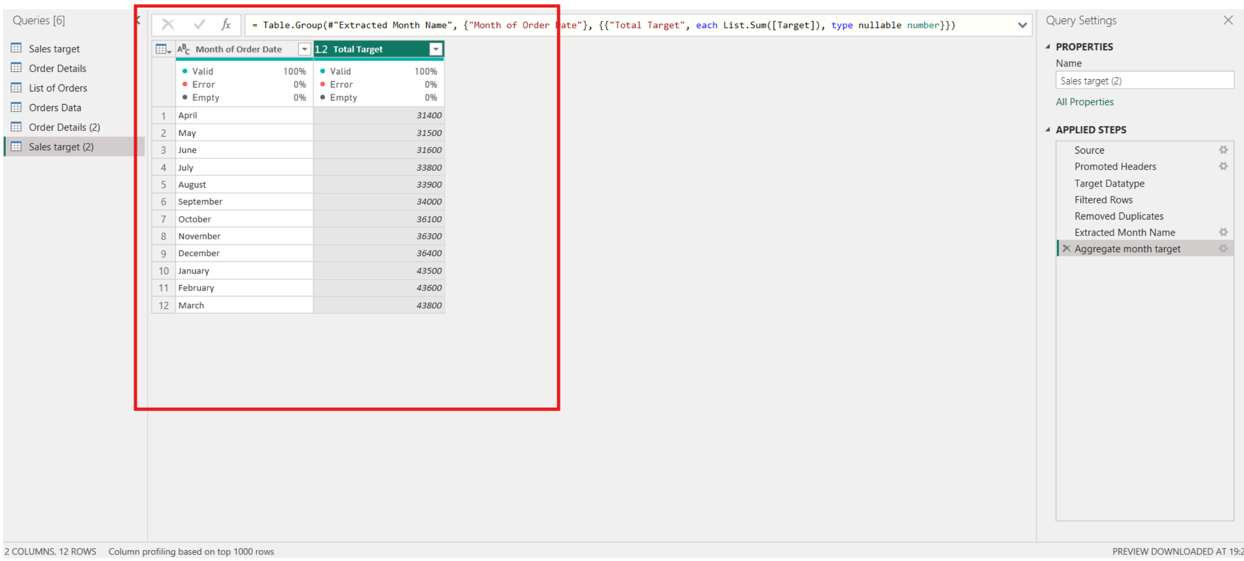


○ Total Amount by Sub-Category



● Duplicate the Sales Target table and aggregate the total Target by Month of order date.



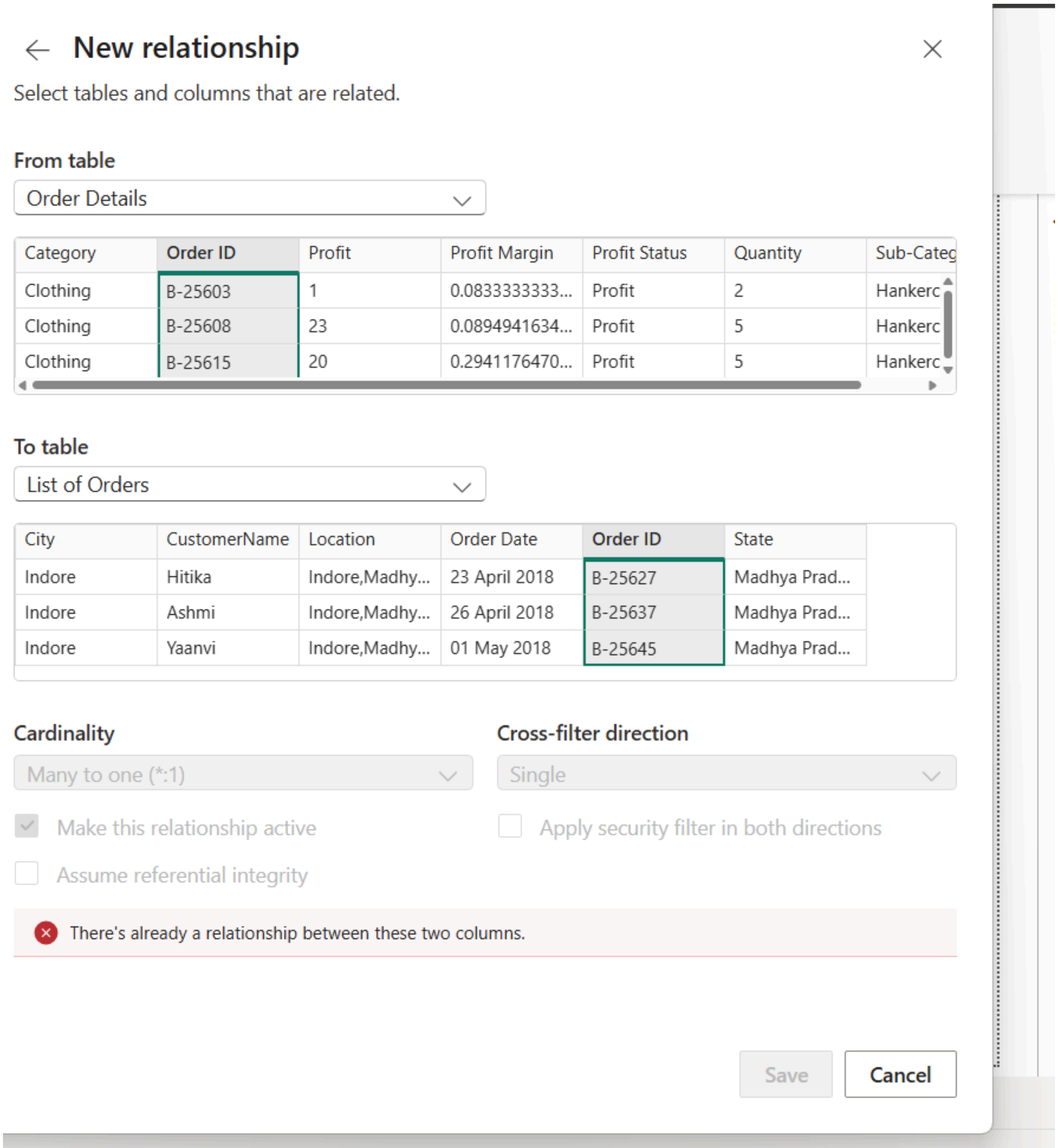


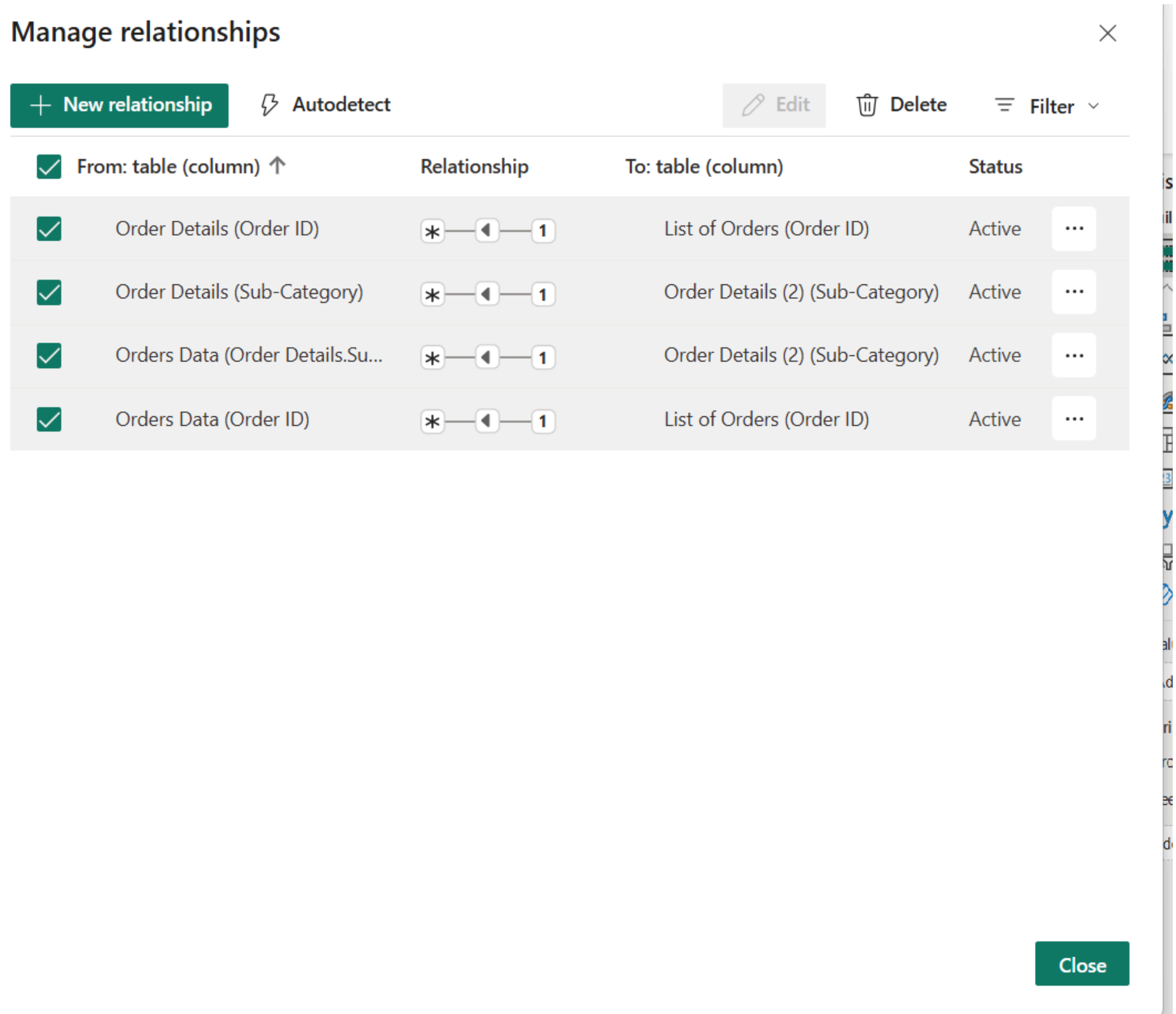
1. Duplicated the Order Details and Sales Target tables for aggregation.
2. Calculated the count of Order IDs to determine the total number of orders.
3. Calculated the average profit for each category.
4. Summed the total amount for each sub-category.
5. Aggregated the total sales target by month of the order date.
6. Grouping and aggregation simplified the data for reporting and analysis.

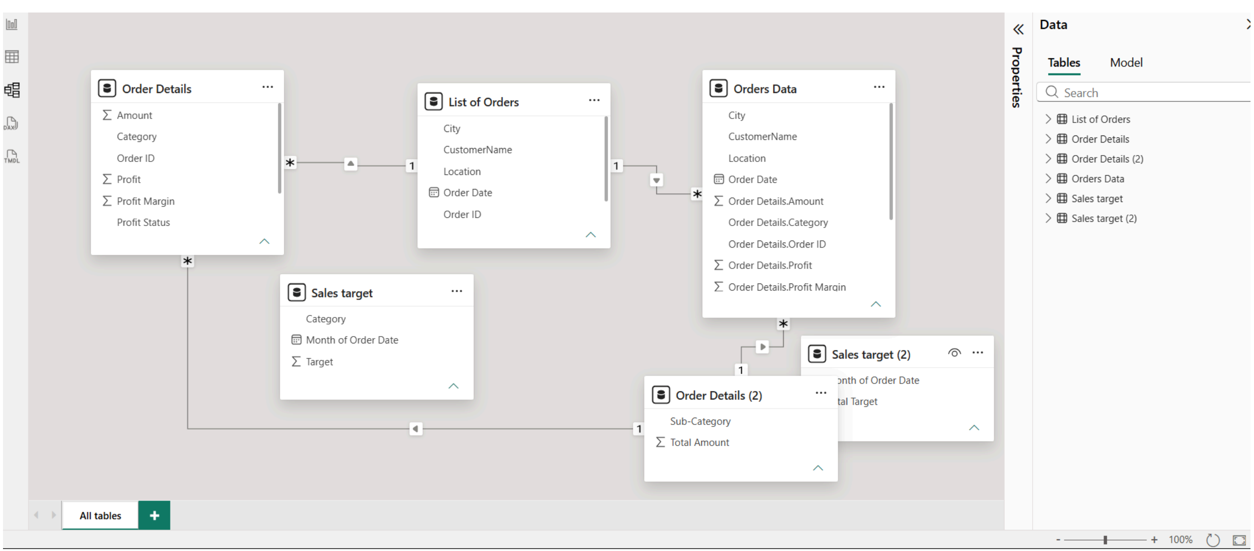
10. Data Modeling:

- Establish a relationship between:
  - List of Orders and Order Details using Order ID
- Establish a relationship between:
  - Order Details and Sales Target using Category

2. Use Manage Relationships to ensure relationships are active and correctly defined.


New relationship
Select tables and columns that are related.
From table
Order Details
Category | Order ID | Profit | Profit Margin | Profit Status | Quantity | Sub-Categ
Clothing | B-25603 | 1 | 0.0833333333... | Profit | 2 | Hankerc
Clothing | B-25608 | 23 | 0.0894941634... | Profit | 5 | Hankerc
Clothing | B-25615 | 20 | 0.2941176470... | Profit | 5 | Hankerc
To table
List of Orders
City | CustomerName | Location | Order Date | Order ID | State
Indore | Hitika | Indore,Madhy... | 23 April 2018 | B-25627 | Madhya Prad...
Indore | Ashmi | Indore,Madhy... | 26 April 2018 | B-25637 | Madhya Prad...
Indore | Yaanvi | Indore,Madhy... | 01 May 2018 | B-25645 | Madhya Prad...
Cardinality
Many to one (*:1)
Cross-filter direction
Single
Make this relationship active
Apply security filter in both directions
Assume referential integrity
There's already a relationship between these two columns.
Save
Cancel


Manage relationships
New relationship
Autodetect
Edit
Delete
Filter
From: table (column)
Relationship
To: table (column)
Status
Order Details (Order ID)
*
1
List of Orders (Order ID)
Active
Order Details (Sub-Category)
*
1
Order Details (2) (Sub-Category)
Active
Orders Data (Order Details.Su...
*
1
Order Details (2) (Sub-Category)
Active
Orders Data (Order ID)
*
1
List of Orders (Order ID)
Active
Close


Order Details
Amount
Category
Order ID
Profit
Profit Margin
Profit Status
List of Orders
City
CustomerName
Location
Order Date
Order ID
Orders Data
City
CustomerName
Location
Order Date
Order Details.Amount
Order Details.Category
Order Details.Order ID
Order Details.Profit
Order Details.Profit Margin
Sales target
Category
Month of Order Date
Target
Sales target (2)
onth of Order Date
tal Target
Order Details (2)
Sub-Category
Total Amount
All tables
Properties
Data
Tables
Model
Search
List of Orders
Order Details
Order Details (2)
Orders Data
Sales target
Sales target (2)
100%

The relationship between List of Orders and Order Details using Order ID was already established.
It was verified in Manage Relationships and found to be active, so no additional relationship creation was required.